

Smart Parking System - Components List

Component	Qty	Technical Description
ESP32 DevKit V1	1	32-bit dual-core microcontroller with integrated Wi-Fi and Bluetooth. Acts as the central controller by processing sensor data, controlling peripherals, and hosting the IoT web dashboard.
HC-SR04 Ultrasonic Sensor	6	Ultrasonic distance sensor operating at 40 kHz with a measurement range of approximately 2–400 cm. Used for real-time parking slot occupancy detection.
IR Obstacle Sensor Module	1	Infrared reflective sensor installed at the entry gate to detect the arrival of a vehicle and trigger automatic parking slot allocation.
SG90 Micro Servo Motor	1	PWM-controlled SG90 micro servo motor used to automate the parking barrier gate by rotating between predefined open and closed positions.
16×2 LCD Display with I2C Module	1	Alphanumeric display used to provide parking instructions, allocated slot information, and system status while minimizing GPIO usage through I2C communication.
74HC595 Shift Register	2	8-bit serial-in, parallel-out shift registers used to expand GPIO outputs for controlling multiple bi-color LEDs using only three ESP32 GPIO pins.
Bi-color LEDs (Red/Green)	6	Two-color LEDs used for visual parking guidance. Different colors and blinking patterns indicate reserved, occupied, and available parking slots.
220 Ω Resistors	12	Current-limiting resistors connected in series with LED terminals to protect the LEDs from excessive current.
Voltage Divider Resistors (1 k Ω & 2 k Ω)	6 pairs	Used on the HC-SR04 Echo lines to reduce the 5 V Echo signal to a safe 3.3 V level compatible with ESP32 GPIO pins.
Breadboard	1	Solderless prototyping board used for power distribution and temporary hardware interconnections during system development.
Jumper Wires	Multiple	Male-to-male, male-to-female, and female-to-female jumper wires used for electrical connections between all hardware modules.

Micro USB Cable	1	Used for programming the ESP32 and supplying power during development and testing.
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